FIITJEE

ALL INDIA TEST SERIES

FULL TEST – VII

JEE (Advanced)-2020

PAPER – 1

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. AC

Sol. $\frac{dN_B}{dt} = \lambda_1 N_A - \lambda_2 N_B$

at, $t = t_0, \frac{dN_B}{dt} = 0, N_B = \frac{3N_0}{2} \Rightarrow 2N_0\lambda_1 e^{-\lambda_1 t_0} - \lambda_0 \frac{3N_0}{2} = 0$

$t_0 = \frac{1}{\lambda_1} \ln \frac{4}{3} \frac{\lambda_1}{\lambda_2}$

2. ABC

Sol. Using work-energy theorem till string become taut

Work done by *qE* = Change in K.E.

$\Rightarrow qEl = \frac{1}{2} mv^2 - 0$

$\Rightarrow v = \sqrt{\frac{2qEl}{2m}}$

When string become taut an impulse is received and speed of the bob becomes

$v\cos 30^o = \sqrt{\frac{3qEl}{2m}}$

Further using work energy theorem speed of the bob becomes

$v' = \sqrt{\frac{5qEl}{2m}}$

$\therefore$ time period $= \frac{2\pi l}{v'} = 2\pi\sqrt{\frac{2ml}{5qE}}$

After 1st half revolution in magnetic field work done by qE =Change in K.E., from lowermost to uppermost point.

$qE.2l = \frac{mv_1^2}{2} - \frac{m}{2}\frac{5qEl}{2m}$

$v_1 = \sqrt{\frac{13qEl}{2m}}$

$\therefore$ Time period of 2nd revolution $= \frac{2\pi l}{v_1} = \pi\sqrt{\frac{2ml}{13qE}}$

$\therefore$ Angular speed of 1st revolution $= \sqrt{\frac{5qE}{2ml}}$

$\therefore$ Angular speed of 2nd revolution $= \sqrt{\frac{13qE}{2ml}}$

$\therefore$ ABC

3. ACD

Sol. Radius of $2m = \frac{r}{3}$, radius of $m = \frac{2r}{3}$. (because C.M of the system should be at rest)

Similarly, velocity of $2m = \frac{v}{3}$; velocity of $m = \frac{2v}{3}$.

$\therefore$ K.E of $2m = \frac{1}{2}m\left(\frac{v}{3}\right)^2$; K.E of $m = \frac{1}{2}m\left(\frac{2v}{3}\right)^2$

$\therefore$ K.E of $2m = \frac{mv^2}{9}$; K.E. of $m = \frac{2mv^2}{9}$.

Time period of $2m = \frac{2\pi(r/3)}{v/3}$; Time period of $m = \frac{2\pi\left(\frac{2\pi}{3}\right)}{2v/3}$

Time period of $2m = \frac{2\pi r}{v}$; Time period of $m = \frac{2\pi r}{v}$

Angular momentum of 2m = $2m\left(\frac{v}{3}\right)\left(\frac{r}{3}\right)$; angular momentum of $m = m\left(\frac{2v}{3}\right)\left(\frac{2r}{3}\right)$

Angular momentum of 2m = $\frac{2mvr}{9}$; angular momentum of $m = \frac{4mvr}{9}$.

4. ABCD

Sol. Direction of magnetic field at P due to $CD = \hat{i} \times \frac{\left(\hat{j}+\hat{k}\right)}{\sqrt{2}} \Rightarrow \frac{k-\hat{j}}{\sqrt{2}}$

Direction of magnetic field at P due to $EF = -\hat{i} \times \frac{\left(\hat{j}-\hat{k}\right)}{\sqrt{2}} \Rightarrow \frac{-k-\hat{j}}{\sqrt{2}}$

Hence direction will be along $-\hat{j}$

Direction of magnetic field at 'O' due to upper frame along $-\hat{i}$

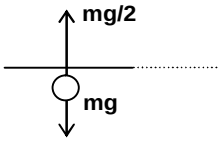
Direction of magnetic field at O due to horizontal frame will be along $-\hat{j}$

5. C

Sol. Travelling harmonic wave is sin or cos function of (ax $\pm$ bt)

6. B

Sol. $ma = mg - \frac{mg}{2} - 6\pi\eta rv$



$ma = 6\pi\eta rv_T - 6\pi\eta rv$

$\frac{dv}{dt} = \frac{6\pi\eta r}{m}(v_T - v)$

$\frac{dv}{v_T - v} = \frac{6\pi\eta r}{m}dt$

$\frac{v_T - v}{v_T} = e^{-\frac{6\pi\eta r}{m}t}$

$v = v_T\left(1 - e^{-\frac{6\pi\eta r}{m}t}\right)$

$v = v_T\left(1 - e^{-\frac{6\pi\eta r}{m}t}\right)$

$\frac{1}{2} = 1 - e^{-\frac{6\pi\eta r}{m}t}$

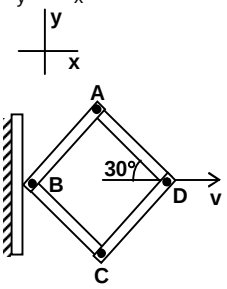
$\frac{6\pi\eta rt}{m} = \ell n2$

$t = \frac{m\ell n2}{6\pi\eta r}$

7. AB

Sol. $v_x = v/2$

$v_y = v_x \cot 30°$



$v_y = \frac{v}{2}\sqrt{3}$

$v_A = v$

8. B

9. C

10. B

Sol. (for Q.8 to 10)

$V_b = i_b R - i_1 R_1 = 0$

$-i_2 R_2 - q/C + + i_1 R_1 = 0$

With $i_2 = \frac{dq}{dt}$

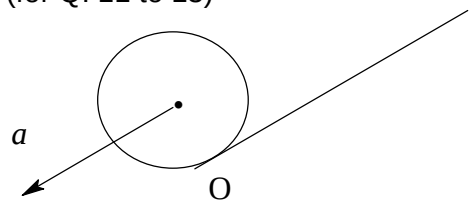
Substituting the given values, we get the result.

11. C

12. A

13. D

Sol. (for Q. 11 to 13)



Taking torques about ICR $\Rightarrow$

$mg \sin\theta \,.\, R = \left(I_{CM} + mR^2\right)\alpha$

$a = \alpha R \Rightarrow \frac{mg \sin\theta R^2}{\left(I_{CM} + mR^2\right)}$

Final $V_{cm} = 0 \therefore t = \frac{V}{a} = \frac{V\left(I_{CM} + mR^2\right)}{mg \sin\theta R^2}$

Also, Applying energy **conservation**:-

$mgh = \frac{1}{2}\left(I_{CM} + mR^2\right)\left(\frac{V}{R}\right)^2$

$\therefore h = \frac{1}{2}\left[\frac{I_{Cm}}{mR^2} + 1\right]\frac{V^2}{g}$

SECTION – B

14. 3

Sol. Flux of magnetic field at any time *t* is

$$\phi = \int_{l}^{2l} \alpha x t l dx = \frac{3}{2}\alpha t l^3$$

$$\therefore |e| = \frac{3}{2}\alpha l^3$$

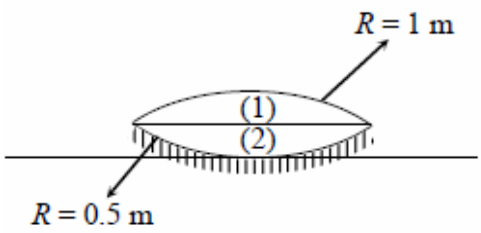
$\therefore$ Charge on capacitor $= \frac{3}{2}C\alpha l^3$

$\therefore \ n = 3$

15. 3

Sol. $\frac{1}{f_1} = (1.5-1)\left(\frac{1}{1}\right) = 0.5m^{-1}$

$$\frac{1}{f_2} = \left(\frac{4}{3}-1\right)\left(\frac{1}{0.5}\right)$$



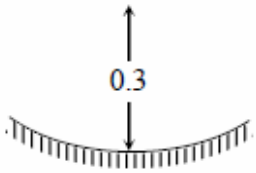
$$= \frac{1}{3}\times\frac{1}{0.5} = \frac{2}{3}m^{-1}$$

Power of the system

$$P = 2P_1 + 2P_2 + P_m$$

$$= 2\times 0.5 + 2\times\frac{2}{3} + \frac{1}{0.25}$$

$$= 1 + \frac{4}{3} + 4 = \frac{3+4+12}{3} = \frac{19}{3}$$



Focal length of equivalent mirror $= -\frac{3}{19}m$

$$\frac{1}{v} + \frac{1}{-0.3} = \frac{1}{-3/19}$$

$$\frac{1}{v} = \frac{1}{0.3} - \frac{19}{3} = \frac{10}{3} - \frac{19}{3} = \frac{-9}{3} = -3$$

$$v = -\frac{1}{3}m$$

$$\Rightarrow d = -\frac{3}{10} + \frac{1}{3} = \frac{1}{30}$$

$$\Rightarrow \frac{1}{30} = \frac{x}{90} \Rightarrow x = 3m$$

16. 2

Sol. Initially let number of molecules of $A = N_A$ and that of $B = N_B$

$$\frac{\lambda N_A}{\lambda N_A + 2\lambda N_B} = \frac{25}{100}$$

$$\therefore 100\lambda N_A = 25\lambda N_A + 50\lambda N_B \Rightarrow N_A = \frac{2}{3}N_B$$

Now, $N_A^{'} = N_A e^{-\lambda t} = \frac{2}{3}N_B e^{-\lambda t}$

$$N_B^{'} = N_B e^{-2\lambda t} = \frac{2}{3}N_B e^{-\lambda t}$$

$$\frac{\frac{2}{3}\lambda N_B e^{-\lambda t}}{\frac{2}{3}\lambda N_B e^{-\lambda t} + 2\lambda N_B e^{-2\lambda t}} = \frac{75}{100} = \frac{3}{4}$$

$$\Rightarrow \frac{1}{1+3e^{-\lambda t}} = \frac{3}{4}$$

$$\Rightarrow 1 = 9e^{-\lambda t}$$

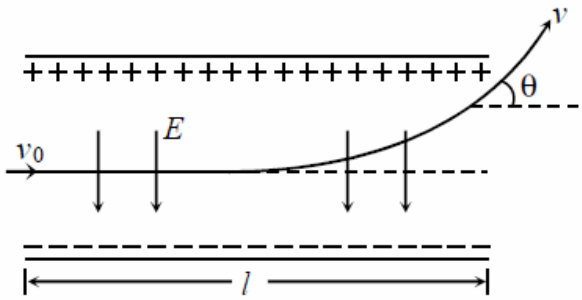
$$\Rightarrow e^{\lambda t} = 9$$

So, $t = \frac{\ln(9)}{\lambda} hr = \frac{2\ln(3)}{\ln(3)} = 2hr$

17. 9

Sol. Since, electron is accelerated through a potential difference *V*, its initial velocity v_0 is given by $\frac{1}{2}mv_0^2 = eV$

Or $v_0 = \sqrt{\frac{2eV}{m}}$ ----------(1)



Since, initial velocity is parallel to plates or normal to the direction of electric field, component of velocity parallel to plates remains constant as v_0.

Hence, time taken by the electron to cross electric field is $t_0 = \frac{l}{v_0}$

Now consider motion of electron, normal to plates.

At some instant *t*, its acceleration = $= \frac{eE}{m} = \frac{eat}{m}$

Let velocity component normal to plates be v_y

$$\therefore \frac{d}{dt} v_y = \frac{eat}{m}$$

$$\therefore \int_0^{v_y} dv = \frac{ea}{m} \int_0^{t_0} t\, dt$$

$$or \;\; v_y = \frac{ea}{2m} t_0^2 = \frac{al^2}{4V} ----------(2)$$

If θ is angular deviation of electron from its initial direction of motion, then pitch of its helical path.

$$p = \frac{2\pi m}{eB} v \cos(90 - \theta)$$

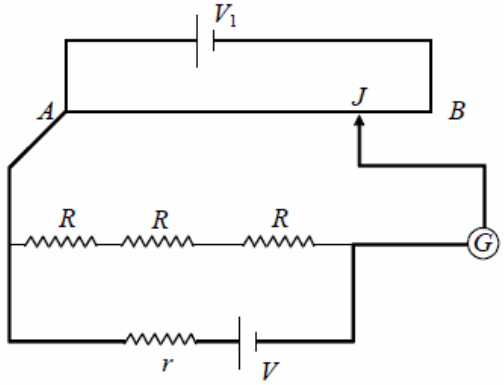
$$\therefore \;\; p = \frac{2\pi m}{eB} v \sin\theta = \frac{2\pi m}{eB} v_y$$

$$or \; p = \frac{\pi m a l^2}{2eBV}$$

$$p = 9mm$$

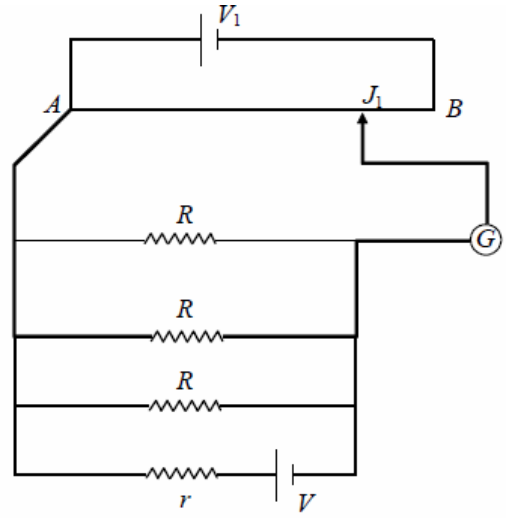
18. 1

Sol. When only switch 2 is closed



$$\frac{3RV}{r+3R} = \frac{V_1}{\lambda L} \lambda L_1 ------(i) \;\; (\lambda\text{-resistance per unit length})$$

When all switches are closed



$$\frac{V}{\frac{R}{3}+r}\times\frac{R}{3}=\frac{V_1}{\lambda L}\lambda L_2 \text{ -------(ii)}$$

Dividing (i) and (ii)

$3R+9r=3r+9R$

$r=R$

$\therefore n=1$

Chemistry

PART – II

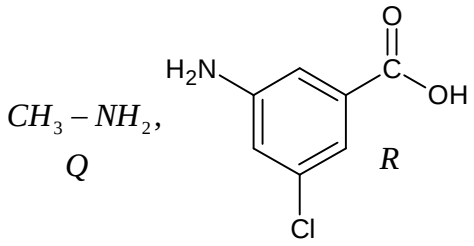
SECTION – A

19. ABC

Sol. On opening the ring, it can convert in to fructose and thus giving Tollen's test.

20. ABD

Sol.



$CH_3 - NH_2$, Q $\qquad$ R

21. ABC

Sol. $s = \left[Zn(OH)_2\right]_{aq} + \left[Zn(OH)\right]^+ + \left[Zn^{2+}\right] + \left[Zn(OH)_3\right]^- + \left[Zn(OH)_4\right]^-$

$$= K_1 + \frac{K_2K_1}{\left[OH^-\right]} + \frac{K_1K_2K_3}{\left[OH^-\right]^2} + K_4\left[OH^-\right]K_1 + K_1K_4K_5\left[OH^-\right]^2$$

22. BCD

Sol. $[A] = [A]_0 - kt = [A]_0 + (\tan 150)t = [A]_0 - \frac{\sqrt{3}}{3}t = 0.5\frac{-\sqrt{3}}{3}t$

Similarly $[C] = [C]_0 - \sqrt{3}t$.....(1)

If $[B] = [A] \Rightarrow t = t_{1/2} = \frac{a}{2k} = \frac{0.5}{2\frac{\sqrt{3}}{3}} = 0.25\sqrt{3}$

Now

$[D] = [C]_0 - [C]$

$= \sqrt{3}t \quad by\ (1)$

$= \sqrt{3} \times 0.25 \times \sqrt{3} = 0.75$

23. CD

Sol. $\left.\begin{matrix} Ti^{3+} \to 3d^1 \\ V^{3+} \to 3d^2 \end{matrix}\right\}$ always have unpaired e^-

24. D

Sol. $\left[Fe(CN)_6\right]^{4-} + NO_2^- + H_2O \longrightarrow \left[Fe(CN)_5\,NO\right]^{2-} + CN^- + 2OH^-$

$\left[Fe(CN)_5\,NO\right]^{2-} + S^{2-} \longrightarrow \left[Fe(CN)_5\,NOS\right]^{4-}$ (purple)

25. AD

Sol. More charge on central atom, more will be CFSE.

26. C

27. D

28. C

29. C

Sol. $I_2 + Na_2S_2O_3 \longrightarrow I^- + Na_2S_4O_6$

30. A

Sol. In an acidic medium MnO_4^- reduces to Mn^{2+}

31. B

Sol. $2MnO_4^- + 5H_2C_2O_4 + 2H^+ \longrightarrow 10CO_2 + 2MnO + 6H_2O$

SECTION – B

32. 0

Sol. $Ag(CN)_2^- \longrightarrow Ag^+ + 2CN^-$

0.1 x 2x

$4\times10^{-17} = \dfrac{4x^3}{0.1}$

$x = 10^{-6}$

Ionic product $= [Ag^+][Cl^-] = 10^{-6}\times0.02 = 2\times10^{-8}$

If $K_{sp} <$ Ionic product

So precipitation will take place.

33. 2

Sol. $0.48\ \mu Ci = 3.7\times10^4\times0.48\,\text{dPs}/m\ell$

For $0.5\,m\ell$ drop, it is $\dfrac{3.7\times10^4\times0.48}{2} dPs$

$\dfrac{444}{60}\text{dPs} = \dfrac{3.7\times10^4\times0.24\times0.1}{V}\text{dPs}$

$V = 120$

So $x = 2$

34. 3

Sol. $E_a = \dfrac{k_1E_{a_1} + k_2E_{a_2}}{k_1 + k_2} = \dfrac{2E_1 + 4E_2}{6} = \dfrac{E_1 + 2E_2}{3}$

$x = 3$

35. 4

Sol. a,b,d,i,m,,n,o,p,q,r

36. 7

Sol. D is wrong other statements are correct

This ore is $FeCr_2O_4$ and it is the prep. Of $K_2Cr_2O_7$ in figure

$FeCr_2O_4 + O_2 + Na_2CO_3 \rightarrow Na_2CrO_4(\text{yellow}) + Fe_2O_3 + CO_2$ after acidification we have

$NaCr_2O_7.2H_2O$ and $Na_2SO_4.10H_2O$

Mathematics

PART – III

SECTION – A

37. AC

Sol. Let $z-3=re^{i\theta}$

$\therefore \quad \frac{z-3}{e^{i\theta}}+\frac{e^{i\theta}}{z-3}=re^{i(\phi-\theta)}+\frac{1}{r}e^{i(\theta+\phi)}$

By solving for imaginary part

$\Rightarrow r-\frac{1}{r}=0$ or $\sin(\phi-\theta)=0$

r = 1

By putting the value we get

$\frac{x-3}{\cos\theta}=\frac{y}{\sin\theta}=r$

38. AD

Sol. $\frac{{}^{50}C_{5r}}{{}^{50}C_{r}}=29\times43\times\frac{a}{b}$

$\Rightarrow \frac{\lfloor r \rfloor 50-r}{\lfloor 5r \rfloor 50-5r}=29\times43\times\frac{a}{b}$

$\Rightarrow 43 \le 50-r \le 46$ as 43 is present in the N_r and a $\notin$ 47 I.

Also $5r \le 25 \cap 50-5r \le 25$ as 29 is present in the D_r

$\Rightarrow r = 5$

39. AB

40. AC

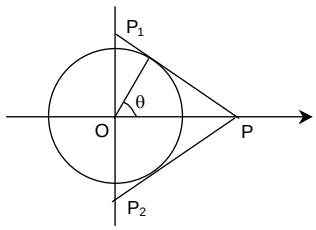
Sol. $OP = 5\sqrt{2}\sec\theta$

$OP_1 = 5\sqrt{2}\,\text{cosec}\theta$

$\Delta PP_1P_2 = \frac{100}{\sin 2\theta}$

$(\Delta \mid PP_1P_2)_{min} = 100 \Rightarrow \theta = \pi/4$

$\Rightarrow OP = 10 \Rightarrow P = (10, 0), (-10, 0)$.



41. BC

Sol. Let $A=\left(3m_1^2,-6m_1\right), B=\left(3m_2^2-6m_2\right), D=\left(3m_3^2,-6m_3\right)$

A and B lies on x + y = 1

$\therefore\ 3m_1^2,-6m_1=0$ and $3m_2^2-6m_2=1$

$3m_1^2-3m_2^2-2(m_1-m_2)=0$

$m_1+m_2=2 \Rightarrow -m_3=2$

$\therefore$ Coordinates of D are (12, 12)

42. ACD

Sol. Equation of tangent at θ

$$\frac{x\cos\theta}{a}+\frac{y\sin\theta}{b}=1 \quad \Rightarrow \quad \frac{x}{a\sec\theta}+\frac{y}{b\,\mathrm{cosec}\,\theta}=1$$

$\therefore$ Intercept $d=\sqrt{a^2\sec^2\theta+b^2\,\mathrm{cosec}^2\theta}$

$$\therefore \quad d^2=a^2\sec^2\theta+b^2\,\mathrm{cosec}^2\theta \Rightarrow \frac{d\left(d^2\right)}{d\theta}=2a^2\frac{\sin\theta}{\cos^3\theta}-2b^2\frac{\cos\theta}{\sin^3\theta}=0$$

i.e $\dfrac{a^2\sin\theta}{\cos^3\theta}=\dfrac{b^2\cos\theta}{\sin^3\theta}$

$\therefore \quad \tan^4\theta=\dfrac{b^2}{a^2}$ $\qquad\qquad$ $\therefore \quad \tan\theta=\pm\sqrt{\dfrac{b}{a}}$

$\theta=\tan^{-1}\sqrt{\dfrac{b}{a}},\ -\tan^{-1}\sqrt{\dfrac{b}{a}}\ \ \theta=\pi-\tan^{-1}\sqrt{\dfrac{b}{a}},-\pi+\tan^{-1}\sqrt{\dfrac{b}{a}}$

43. ABC

Sol. The given hyperbola is a rectangular hyperbola so Δ CEF is right angle and eccentricity of the hyperbola is $\sqrt{2}$

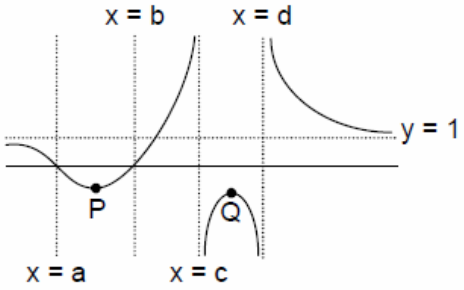
44. B

45. B

46. B

Sol. (for Q. 44 to 46)

We cannot be sure about the coordinates of P and Q as they depend on a, b, c and d



47. D

48. D

49. C

Sol. (for Q. 47 to 49)

$$\int_0^x\sqrt{1-\left(f''(t)\right)^2}\,dt=\int_0^x f'(t)\,dt \Rightarrow \sqrt{1-\left(f''(x)\right)^2}=f'(x) \Rightarrow f''(x)=\pm\sqrt{1-\left(f'(x)\right)^2}$$

$$\Rightarrow x=\pm\sin^{-1}f'(x) \Rightarrow f(x)=\pm(1-\cos x)=\pm 2\sin^2\frac{x}{2}$$

$$f(x) \le |f(x)| = 2\sin^2\frac{x}{2} < 2\cdot\frac{x^2}{4} = \frac{x^2}{2}$$

$$\int_0^1 f(x)\,dx \le \int_0^1 |f(x)|\,dx = \int_0^1 2\sin^2\frac{x}{2}\,dx < \frac{1}{2}\int_0^1 x^2\,dx = \frac{1}{6}$$

SECTION – B

50. 2

Sol. $x^2 + px + 2 = 0 < {}^{r}_{s}$ and $x^2 + rx + s = 0 < {}^{p}_{q}$

$r + s = -p \quad (1)$

$rs = q \quad (2)$

$p + q = -r \quad (3)$

$pq = s \quad (4)$

$(2)\times(4) \Rightarrow pqrs = qs \Rightarrow pr = 1 \quad (5)$

$(1)+(3) \Rightarrow p+q+r+s = -p-r$

$\Rightarrow 2(p+r)+q+s = 0 \quad (6)$

Now p is a root of $x^2 + rx + s = 0 \Rightarrow p^2 + rp + s = 0$

And r is a root of $x^2 + px + q = 0 \Rightarrow r^2 + pr + q = 0$

Adding those two

$\Rightarrow (p+r)^2 - 2(p+r) = 0 \Rightarrow p+r = 0, p+r = 2$

Also $pr = 1 \Rightarrow p = 1, r = 1$

From $(4) pq = s \Rightarrow q = s$

$\therefore q + s = -2(p+r)$

$\Rightarrow q + s = -4 \Rightarrow q = s = -2$

$\therefore 4\times 1 - 2 + 2 - 2 = 2.$

51. 6

Sol. $5^{\log_{1/5}(1/2)} + \log_{\sqrt{2}}\frac{4}{\sqrt{7}+\sqrt{3}} + \log_{1/2}\frac{1}{10+2\sqrt{21}}$

$= 5^{\log_5 2} + 2\log_2(\sqrt{7}-\sqrt{3}) + \log_2(10+2\sqrt{21})$

$=6$

52. 7

Sol. $\sum_{r=1}^{n}\left[(r+1)^{P+1} - r^{P+1}\right] = (P+1)\sum_{r=1}^{n} r^P + {}^{P+1}C_2\sum_{r=1}^{n} r^{P-1} + \ldots\ldots + \sum_{r=1}^{n}(1)$

$a_0 = \frac{1}{P+1}$

53. 3

Sol. $\because \frac{d}{dx}\left\{\frac{dy}{dx}.y\right\} = e^x$

$\Rightarrow y\frac{dy}{dx}=e^x+c$

$\Rightarrow 1\times 1=e^0+c\Rightarrow c=0$

$\Rightarrow tdy=e^x dx$

$\frac{1}{2}y^2=e^x+c\therefore\frac{1}{2}=1-c\Rightarrow c=-\frac{1}{2}$

$y^2=2e^x-1\Rightarrow f^2(x)=2e^x-1$

54. 7

Sol. Let $x+3=X$ & $y-1=Y$

$\therefore \frac{dY}{dX}=\frac{(X+Y)^2-Y^2}{X^2}=\frac{X^2+2XY}{X^2}$

$\Rightarrow \frac{dY}{dX}-\frac{2Y}{X}=1$

I.F. = $e^{\int-\frac{2}{X}dX}=e^{-2\ln X}=\frac{1}{X^2}$

$\therefore \quad Y\times\frac{1}{X^2}=\int\frac{1}{X^2}\times 1\,dX+c$

$\Rightarrow \frac{(y-1)}{(x+3)^2}=-\frac{1}{(x+3)}+c \quad \Rightarrow (y-1)=c(x+3)^2-(x+3)$

$\therefore 0=9c-3$

$\Rightarrow c=\frac{1}{3}$

$y=1+\frac{1}{3}(x+3)^2-(x+3)$

$y(3)=1+12-6=7$